

BRAC University

Department of Mathematics and Physical Sciences

MAT110

Lecture 17

**Transformation of Coordinates
Rotation of Axes and Conic Classification**

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1. General Equation of the Second Degree

General Equation

The general equation of the second degree in x and y is

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0.$$

Its nature is determined by the coefficients a , h , b and, in the degenerate case, by the determinant Δ .

Nature of the Conic

The equation

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

represents the following conics:

- (i) a **circle** if $a = b$ and $h = 0$;
- (ii) an **ellipse** if $ab - h^2 > 0$;
- (iii) a **hyperbola** if $ab - h^2 < 0$;
- (iv) a **parabola** if $ab - h^2 = 0$;
- (v) a **pair of straight lines** if $\Delta = 0$.

Examples of Conic Classification

(i) Circle

$$x^2 + y^2 = 5^2, \quad x^2 + y^2 - 2gx - 4y - 5 = 0.$$

In each equation, $a = b$ and $h = 0$.

(ii) Ellipse

$$144x^2 + 25y^2 = 1, \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Also,

$$3x^2 + 2xy + 5y^2 - 2x - 8y - 8 = 0$$

has $a = 3$, $b = 5$, and $h = 1$. Hence

$$ab - h^2 = 15 - 1 = 14 > 0.$$

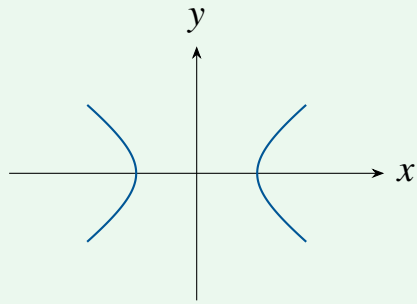
(iii) Hyperbola

$$x^2 - y^2 = 1.$$

For

$$9x^2 + 24xy + 2y^2 - 6x + 20y + 41 = 0,$$

we have $a = 9$, $b = 2$, and $h = 12$, so $ab - h^2 < 0$.

**(iv) Parabola**

$$y^2 = 4(2)x.$$

For

$$x^2 - 2xy + y^2 - 2x + 3y - 8 = 0,$$

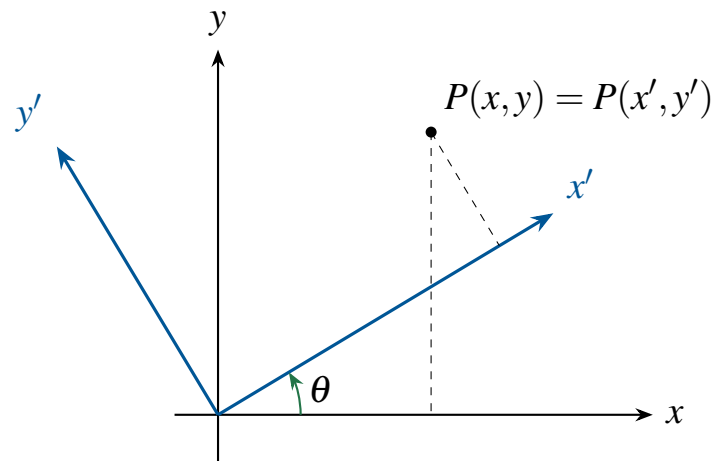
we have $a = 1$, $b = 1$, and $h = -1$, so $ab - h^2 = 0$.

(v) Pair of Straight Lines

$$x^2 - y^2 = 0, \quad x^2 + 2xy + y^2 = 0.$$

2. Rotation Transformation

Suppose the rectangular axes x' and y' are obtained by rotating the original axes x and y counterclockwise through an angle θ .



Rotation Formulas

The coordinates are related by

$$\begin{aligned}x &= x' \cos \theta - y' \sin \theta, \\y &= x' \sin \theta + y' \cos \theta.\end{aligned}$$

2.1. Effect on the General Equation of the Second Degree

Consider

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0. \quad (1)$$

Using the rotation formulas in (1), we obtain

$$\begin{aligned}&a(x' \cos \theta - y' \sin \theta)^2 \\&+ 2h(x' \cos \theta - y' \sin \theta)(x' \sin \theta + y' \cos \theta) \\&+ b(x' \sin \theta + y' \cos \theta)^2 \\&+ 2g(x' \cos \theta - y' \sin \theta) + 2f(x' \sin \theta + y' \cos \theta) + c = 0.\end{aligned} \quad (2)$$

Let (2) be simplified to

$$a'x'^2 + 2h'x'y' + b'y'^2 + 2g'x' + 2f'y' + c' = 0. \quad (3)$$

Comparing (3) with (2),

$$a' = \frac{a+b}{2} + \frac{a-b}{2} \cos 2\theta + h \sin 2\theta,$$

$$\begin{aligned}
 b' &= \frac{a+b}{2} - \frac{a-b}{2} \cos 2\theta - h \sin 2\theta, \\
 h' &= h \cos 2\theta - \frac{a-b}{2} \sin 2\theta, \\
 g' &= g \cos \theta + f \sin \theta, \\
 f' &= -g \sin \theta + f \cos \theta, \\
 c' &= c.
 \end{aligned}$$

2.2. Invariant Properties

Invariants under Rotation Transformation

Under a rotation of axes,

- (i) $a' + b' = a + b$;
- (ii) $a'b' - h'^2 = ab - h^2$;
- (iii) $\Delta' = \Delta$;
- (iv) $c' = c$.

2.3. Removal of the $x'y'$ Term

If we want to remove the $x'y'$ term from the transformed equation, then $h' = 0$. Therefore,

$$\begin{aligned}
 h \cos 2\theta - \frac{a-b}{2} \sin 2\theta &= 0, \\
 \frac{a-b}{2} \sin 2\theta &= h \cos 2\theta, \\
 \tan 2\theta &= \frac{2h}{a-b}.
 \end{aligned}$$

Hence the required angle is

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{2h}{a-b} \right).$$

3. Transformation of a Conic

Example

Decide the nature of the conic

$$9x^2 + 24xy + 2y^2 - 6x + 20y + 41 = 0,$$

and transform the equation so as to remove the x , y , and xy terms.

Answer

For the given equation,

$$a = 9, \quad b = 2, \quad h = 12, \quad g = -3, \quad f = 10, \quad c = 41.$$

Now,

$$ab - h^2 = 9(2) - 12^2 = -126 < 0.$$

Therefore, the conic is a **hyperbola**.

Part 01: Removing the x and y terms using a translation

Let

$$F(x, y) = 9x^2 + 24xy + 2y^2 - 6x + 20y + 41. \quad (4)$$

To remove the linear terms, translate the origin to a point (α, β) where

$$\frac{\partial F}{\partial x} = 0 \quad \text{and} \quad \frac{\partial F}{\partial y} = 0.$$

Here,

$$\begin{aligned} \frac{\partial F}{\partial x} &= 18x + 24y - 6 = 0, \\ \frac{\partial F}{\partial y} &= 24x + 4y + 20 = 0. \end{aligned}$$

Solving these equations gives

$$\alpha = -1, \quad \beta = 1.$$

Thus, the new origin is $(-1, 1)$, and the translation is

$$x = x' - 1, \quad y = y' + 1.$$

After translation, the equation takes the form

$$ax'^2 + 2hx'y' + by'^2 + F(\alpha, \beta) = 0.$$

Since

$$F(-1, 1) = 54,$$

the translated equation is

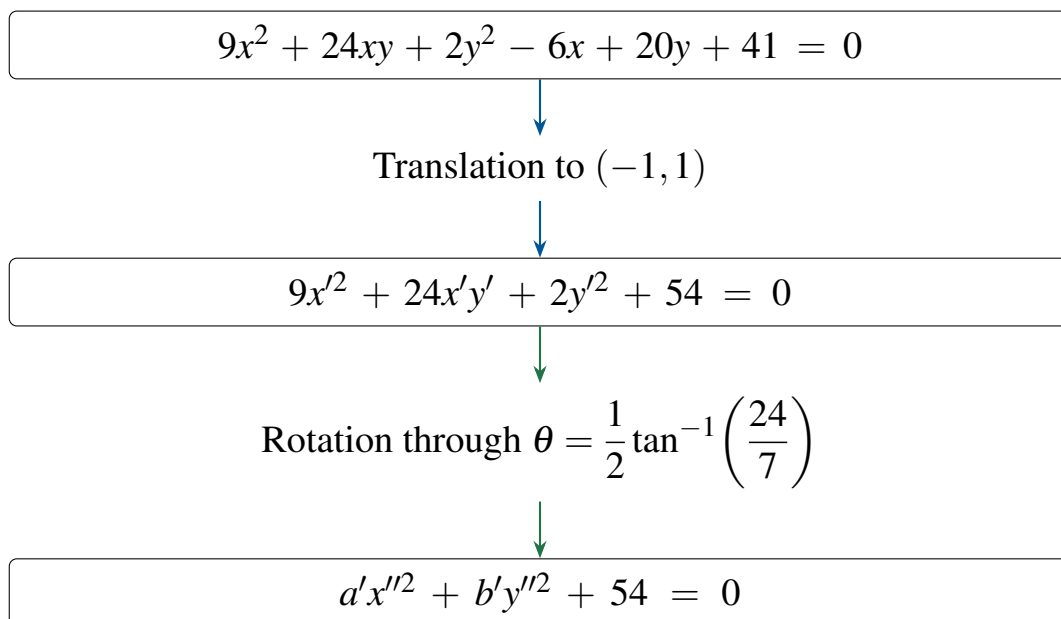
$$9x'^2 + 24x'y' + 2y'^2 + 54 = 0. \quad (5)$$

Part 02: Removing the $x'y'$ term using a rotation

For equation (5),

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{2h}{a-b} \right) = \frac{1}{2} \tan^{-1} \left(\frac{24}{7} \right).$$

The transformation proceeds as follows.



Using the invariant properties,

$$a' + b' = a + b = 9 + 2 = 11, \quad (6)$$

and, since the mixed term has been removed,

$$a'b' = ab - h^2 = 9(2) - 12^2 = -126. \quad (7)$$

Therefore,

$$\begin{aligned} (a' - b')^2 &= (a' + b')^2 - 4a'b' \\ &= 11^2 - 4(-126) \\ &= 625. \end{aligned}$$

Thus,

$$a' - b' = \pm 25. \quad (8)$$

Case (+): $a' - b' = 25$

Together with $a' + b' = 11$, this gives

$$2a' = 36, \quad a' = 18, \quad b' = -7.$$

Hence,

$$\boxed{18x''^2 - 7y''^2 + 54 = 0}. \quad (9)$$

Case (-): $a' - b' = -25$

Together with $a' + b' = 11$, this gives

$$2a' = -14, \quad a' = -7, \quad b' = 18.$$

Hence,

$$\boxed{-7x''^2 + 18y''^2 + 54 = 0}. \quad (10)$$

The two final equations differ only by an interchange of the rotated axes.